

Asterios Tsiourvas

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EDUCATION

Massachusetts Institute of Technology (MIT), Cambridge, MA, US

Aug 2020 - Sep 2025

PhD in Operations Research (GPA 5.0/5.0, top of class).

- Research in Deep Learning & Optimization. *Advisor:* Prof. Georgia Perakis.
- *Relevant Coursework:* Linear, Integer, Combinatorial & Robust Optimization, Machine Learning & Optimization, Statistical Learning, Fundamentals of Probability, Deep Learning Theory, Natural Language Processing.

National Technical University of Athens (NTUA), Athens, Greece

Sep 2019 - Jul 2020

MSc in Data Science & Machine Learning. Ranked 1st out of 15 graduate students (GPA: 9.94/10)

- Research in Machine Learning & Optimization.
- *Relevant Coursework:* Machine Learning, Deep Learning, Data Mining, Stochastic & Statistical Optimization, Convex Optimization, Algorithmic Data Science.

National Technical University of Athens (NTUA), Athens, Greece

Sep 2014 - Oct 2019

BSc & MSc in Electrical and Computer Engineering. Ranked 1st out of 267 graduate students (GPA: 9.76/10)

- Major in Computer Science & Minor in Mathematics.
- *Relevant Coursework:* Algorithms, Artificial Intelligence, Neural Networks, Software Engineering, Parallel & Distributed Systems, Advanced Databases, Stochastic Processes, Queuing Systems, Information Theory.

EXPERIENCE

Quantitative Researcher, Jump Trading, New York, NY

Sep 2025 -

- Conduct research on advanced AI, machine learning, and optimization methods for large-scale, real-time decision-making in financial markets using high-frequency data.
- Design scalable predictive and prescriptive algorithms and software systems for ultra-low-latency, risk-aware decision processes with rigorous validation and performance analysis.
- Translate research insights into production-ready, decision-focused AI frameworks that enhance the robustness, efficiency, and resilience of market operations.

Research Assistant, MIT Operations Research Center, Cambridge, MA

Aug 2020 - May 2025

- Research in the intersection of deep learning and optimization with applications in neural network optimization, causal ML, counterfactual explanations, hierarchical time series, LLMs, and healthcare. Co-authored and published 15 papers in major conferences (ICML, AISTATS, KDD) and journals (PNAS, MSOM, POM) (700+ citations, Google Scholar).
- Collaborators: (i) MIT-IBM Watson AI Lab, research on hierarchical time-series and causal LLM routing & prompt optimization, (ii) Center for Disease Control and Prevention (CDC), research on nationwide prediction of COVID-19 related cases and deaths and (iii) UMass Memorial Medical Center, research on optimal patient scheduling to ED beds.

Quantitative Research Intern, Jump Trading, New York, NY

June 2024 - Aug 2024

- Conducted research on quantitative modeling and decision-making methods for financial markets, designing and evaluating predictive and prescriptive models using large-scale historical and high-frequency data.
- Performed empirical analysis and systematic backtesting to assess robustness, validate assumptions, quantify risk and uncertainty, and translate findings into actionable algorithmic frameworks for expert teams.

Research Intern, IBM Research, Yorktown Heights, NY

May 2023 - Aug 2023

- Conducted research on algorithms for generating realistic counterfactual explanations from deep learning models.
- Co-authored a research paper published in AISTATS 2024 and filed a US patent.

Research Intern, IBM Research, Yorktown Heights, NY

May 2022 - Aug 2022

- Conducted research in the intersection of deep learning and causal inference.
- Co-authored a research paper published in ICML 2023 and filed two US Patents.

Research Assistant, NTUA Computing Systems Laboratory

Mar 2019 - Aug 2020

- Developed a distributed machine learning platform to forecast energy usage of telecommunication base stations for the EU research project BigOptiBase, enabling real-time insights for energy footprint reduction.
- Research on optimizing Amazon EC2 Spot Instances System by developing a dynamic resource allocation and pricing system for revenue maximization.
- Co-authored two research papers published in IEEE Big Data 2019 and IEEE Cloud 2021.

Business Intelligence Intern, Oracle, Athens, Greece

July 2018 - Aug 2018

- Interned as a software engineer on a data warehousing project.

PUBLICATIONS

Journal Articles

- **Optimizing Objective Functions from Trained ReLU Neural Networks**
Co-author: G. Perakis. Under major revision at *Management Science*. 2022 INFORMS RMP Spotlight Paper.
- **A Granular View of the Emergency Department Length of Stay: Improving Predictive Power and Extracting Actionable Insights**
Co-authors: M. Canellas, D. Pachamanova, G. Perakis, O. Skali Lami. *Annals of Emergency Medicine*.
- **A Granular Approach to Optimal and Fair Patient Placement in Hospital Emergency Departments**
Co-authors: M. Canellas, D. Pachamanova, G. Perakis, O. Skali Lami. *Production and Operations Management*.
- **COVID-19: Prediction, prevalence, and the operations of vaccine allocation**
Co-authors: G. Perakis et al. *Manufacturing and Service Operations Management (MSOM)*.
- **Discrete Gradient Flow Approximations of High Dimensional Evolution Partial Differential Equations via Deep Neural Networks**
Co-authors: E. Georgoulis, M. Loulakis. *Communications in Nonlinear Science and Numerical Simulation*.
- **The United States COVID-19 Forecast Hub Dataset**
Co-authors: E. Cramer et al. *Scientific Data - Nature*.
- **Evaluation of individual and ensemble probabilistic forecasts of COVID-19 mortality in the US**
Co-authors: E. Cramer et al. *Proceedings of the National Academy of Sciences (PNAS)*.

Conference Proceedings

- **Causal LLM Routing: End-to-End Regret Minimization from Observational Data**
Co-author: W. Sun, G. Perakis. *Conference on Neural Information Processing Systems (NeurIPS) 2025*.
- **Overcoming the Optimizer's Curse: Obtaining Realistic Prescriptions from ReLU Neural Networks**
Co-author: G. Perakis. *International Conference on Machine Learning (ICML) 2024*.
- **Learning the Optimal Reconciliation for Hierarchical Time Series**
Co-authors: G. Perakis, W. Sun, P. Y. Chen, Y. Zhu. *International Conference on Machine Learning (ICML) 2024*.
- **Manifold-Aligned Counterfactual Explanations for Neural Networks**
Co-authors: W. Sun, G. Perakis. *International Conference on Artificial Intelligence and Statistics (AISTATS) 2024*.
- **Neural-Informed Decision Trees**
Co-author: G. Perakis. *ACM SIGKDD Conference on Knowledge Discovery and Data Mining 2023. Workshop on Ethical Artificial Intelligence*.
- **Learning Prescriptive ReLU Networks**
Co-author: W. Sun. *International Conference on Machine Learning (ICML) 2023*.

- **A Mechanism Design and Learning Approach for Revenue Maximization on Cloud Spot Markets**
Co-authors: C. Bitsakos, I. Konstantinou, D. Fotakis, N. Koziris. *IEEE International Conference on Cloud Computing 2021*.
- **The Power of Analytics in Epidemiology for COVID-19: Prediction, Prevalence and Vaccine Allocation**
Co-authors: G. Perakis et al. *INFORMS Conference on Service Science 2021*.
- **BigOptiBase: Big Data Analytics for Base Station Energy Consumption Optimization**
Co-authors: E. Kassela, N. Provatas, I. Konstantinou and N. Koziris. *IEEE International Conference on Big Data 2019*.

TALKS

- **Causal LLM Routing: End-to-End Regret Minimization from Observational Data**
Greeks in AI 2025 Symposium, NTUA - Computer Systems Lab (CSLab), Invited Talk, 2025.
- **Overcoming the Optimizer’s Curse: Obtaining Realistic Prescriptions from ReLU Neural Networks**
International Conference on Machine Learning (ICML) 2024, INFORMS Annual Meeting 2024 & POMS 2024.
- **Learning the Optimal Reconciliation for Hierarchical Time Series**
International Conference on Machine Learning (ICML) 2024.
- **Manifold-Aligned Counterfactual Explanations for Neural Networks**
International Conference on Artificial Intelligence and Statistics (AISTATS) 2024.
- **Learning Prescriptive ReLU Networks**
EPFL - Signal Processing Lab (LTS4) Group, Invited Talk, 2023.
- **Neural-Informed Decision Trees**
INFORMS MSOM Conference 2023, ACM SIGKDD 2023.
- **Learning Prescriptive ReLU Networks**
International Conference on Machine Learning (ICML) 2023.
- **Optimizing Objective Functions from ReLU Neural Networks via Sampling**
INFORMS Annual Meeting 2022 & 2022 INFORMS RMP Conference (Spotlight Session).
- **A Granular Approach to Optimal and Fair Patient Placement in Hospital Emergency Departments**
INFORMS MSOM Conference 2022.
- **A Mechanism Design & Learning Approach for Revenue Maximization on Cloud Spot Markets**
IEEE International Conference on Cloud Computing 2021.

PROFESSIONAL SERVICE

- **Program Committee Member (Reviewer)**, *AAAI Conference on Artificial Intelligence (AAAI) 2026* — Reviewed 3 research papers. (AAAI is one of the premier international conferences in artificial intelligence.)
- **Peer Reviewer**, *Conference on Neural Information Processing Systems (NeurIPS) 2025* — Reviewed 5 research papers. (NeurIPS is one of the most selective and prestigious AI conferences worldwide.)
- **Peer Reviewer**, *International Conference on Learning Representations (ICLR) 2026* — Reviewed 5 research papers. (ICLR is one of the most selective and prestigious AI conferences worldwide.)
- **Peer Reviewer**, *International Conference on Artificial Intelligence and Statistics (AISTATS) 2024, 2026* — Reviewed 3, 3 research papers. (The leading venue in machine learning and statistics.)
- **Peer Reviewer**, *Transactions on Machine Learning Research (TMLR)* — Reviewed 7 research papers. (A highly regarded open-access journal publishing cutting-edge research in machine learning.)
- **Peer Reviewer**, *International Transactions in Operational Research* — Reviewed 2 journal manuscripts. (An international scholarly journal recognized in the fields of operations research and analytics.)
- **Peer Reviewer**, *INFORMS Journal on Data Science* — Reviewed 1 journal manuscript. (An international scholarly journal focusing on data science, machine learning, and their applications in operations research and analytics.)
- **Grant Proposal Reviewer**, *MIT-IBM Watson AI Lab* — Evaluated a research funding proposal. (The MIT-IBM Lab is a high-profile industry-academia partnership funding cutting-edge AI research.)

US PATENTS

- **Counterfactual Prediction and Interpretable Policy Learning from Observational Data using Prescriptive ReLU Networks**
A. Tsiourvas, M. Ettl & W. Sun. US Patent App. 18/217091. Status: Published. Publication No: 20250005347.
- **Manifold-Aligned Counterfactual Explanations**
A. Tsiourvas, M. Ettl & W. Sun. US Patent App. 18/618596. Status: Published. Publication No: 20250307599.
- **Hierarchical Time Series Forecasting**
A. Tsiourvas, G. Perakis, P.Y. Chen, W. Sun & W. Y. Zhu. US Patent App. 18/984896. Status: Filed.
- **An Integrated Ssystem for LLM Routing with Observational Data**
A. Tsiourvas, W. Sun, G. Perakis. Under preparation. P202501674US01 (IBMP1446US).

HONORS AND AWARDS

- **Nominee for Full Membership, Sigma Xi, The Scientific Research Honor Society** (2025–Present).
Nominated in recognition of scholarly achievements and contributions to scientific knowledge. Sigma Xi is one of the oldest and most prestigious scientific honor societies, with membership granted solely through peer nomination.
- **Theodore Vassilakis Fellow** (2023–2024).
Highly competitive fellowship awarded by MIT Sloan School of Management for exceptional academic record and research contributions.
- **MIT GSC Conference Grant** (2023).
Highly competitive travel grant from the MIT Graduate Student Council to present research at the International Conference on Machine Learning (ICML 2023).
- **Spotlight Presentation, INFORMS Revenue Management and Pricing Section Conference** (2022).
One of a select group of presentations chosen for spotlight status for the paper *Optimizing Objective Functions from Trained ReLU Neural Networks*.
- **INFORMS Early Career Professionals Network (ECPN) Member** (2022).
Selected as one of only 20 operations research graduate students nationwide for this competitive early-career cohort.
- **2nd Place, INFORMS Doing Good with OR Competition** (2021).
National award for *The Power of Analytics in Epidemiology for COVID-19: Prediction, Prevalence and Vaccine Allocation*, recognizing impactful OR applications to societal challenges.
- **Honorable Mention, INFORMS Public Sector OR Best Paper Award** (2021).
For the same COVID-19 research, awarded by the INFORMS Public Sector OR Section in a competitive peer-reviewed process.
- **Best Paper Award, INFORMS Conference on Service Science** (2021).
For outstanding research contribution in service science for the COVID-19 epidemiological analytics project.
- **Onassis Foundation Graduate Fellowship** (2021–2025).
Prestigious fellowship awarded annually to the most outstanding Greek PhD students worldwide.
- **Foundation for Education and European Culture Graduate Fellowship** (2021–2025).
Merit-based fellowship recognizing the top-performing Hellenic PhD students globally.
- **Gerondelis Foundation Graduate Fellowship** (2021–2022).
Competitive U.S.-based fellowship for exceptional graduate students of Greek nationality.
- **Ministry of Education Excellence Award** (2018–2019).
Greek national award for graduating 1st in rank in Electrical and Computer Engineering.
- **Tzafestas Excellence Award** (2018–2019).
NTUA award for graduating 1st in rank in Electrical and Computer Engineering.
- **Chrisovergis Excellence Award** (2018–2019).
NTUA award for graduating 1st in rank in Electrical and Computer Engineering.
- **Kontaxis Excellence Award** (2018–2019).
NTUA award for graduating 1st in rank in Computer Science within Electrical and Computer Engineering.

- **Papakiriakopoulos Excellence Award in Mathematics** (2014–2016).
NTUA award for ranking 1st in Mathematics in Electrical and Computer Engineering.
- **Kritikos Excellence Award in Mathematics** (2014–2015).
NTUA award for ranking 1st in Mathematics in Electrical and Computer Engineering.

MEDIA COVERAGE

- **16 Posters You Can’t Miss at ICML 2024**, MIT Jameel Clinic (2024). Independent coverage highlighting my invited presentation at ICML 2024, one of the premier conferences in machine learning. [Link]
- **Granular Approach to Optimal and Fair Emergency Department Operations**, MIT Jameel Clinic (2024). Public recognition on the official MIT Jameel Clinic LinkedIn page for my research on equitable and efficient emergency department management. [Link]
- **Overcoming the Optimizer’s Curse: Obtaining Realistic Prescriptions from Neural Networks**, MIT Jameel Clinic (2024). Independent feature by MIT Jameel Clinic recognizing my contribution to advancing optimization in deep learning. [Link]
- **Learning Optimal Projection for Forecast Reconciliation of Hierarchical Time Series**, MIT Jameel Clinic (2024). MIT Jameel Clinic research spotlight featuring my work as lead author on novel methods for hierarchical time series forecasting. [Link]
- **Getting Treated in the ED Painlessly is Possible**, MIT Sloan Health Systems Initiative (2023). Independent coverage by MIT Sloan highlighting my role as first author in research improving patient experience in emergency departments. [Link]
- **Research Update: Improving Hospital Operations**, MIT Sloan Health Systems Initiative (2023). Independent reporting by MIT Sloan recognizing my first-author research contributions to optimizing hospital operations. [Link]
- **COVID-19 Forecast Hub: MIT-Cassandra Model**, U.S. Centers for Disease Control and Prevention (2020–2023).
Contributed the MIT-Cassandra ensemble to the CDC’s national COVID-19 ensemble forecasts. This model was consistently among the *best-performing* methods in accuracy and reliability, and it informed high-stakes decision-making on resource allocation and pandemic response. See documentation on the Forecast Hub Community page (link) and the Zoltar model profile (link).
- **COVID-19 Prediction, Prevalence, and the Operations of Vaccine Allocation**, MIT Jameel Clinic (2022). MIT Jameel Clinic coverage of my research addressing COVID-19 prevalence prediction and optimal vaccine allocation strategies. [Link]
- **PROMITHEAS T20**, National Technical University of Athens Newspaper (2022). Featured my master’s thesis in the first research work presented on my undergraduate university’s transition to the AI era; published in the official newspaper of NTUA (in Greek). [Link]
- **Onassis Foundation Annual Scholarships**, Proto Thema Newspaper (2021). National media recognition in Greece’s most widely circulated newspaper for receiving a prestigious Onassis Foundation scholarship (in Greek). [Link]

TECHNICAL SKILLS

- **Programming**
Python, Julia, R, C/C++, Java, Matlab.
- **Software Tools**
PyTorch, HuggingFace, LangChain, Transformers, TensorFlow, Scikit-Learn, NumPy, Pandas, SQL.

LANGUAGES

Greek (native) **English** (C2, TOEFL 111/120) **French** (B2)